Highlights from two astronomy problems for Lagrange's equations: the cosmological redshift and the Lagrange points

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These notes use only what is in Taylor, *Classical Mechanics*, Sections 7.5, 7.6 and 7.8: Lagrange's equations, a coordinate change that depends on time, ignorable coordinates, and "no t in $\mathcal{L}$ gives a conserved Hamiltonian." The cosmology half follows the standard textbook route for a particle in an expanding universe (P. J. E. Peebles, *The Large-Scale Structure of the Universe*, Princeton, 1980, Chapter II); the second half is the restricted three-body problem as every celestial-mechanics book sets it up (e.g. Murray and Dermott, *Solar System Dynamics*, Cambridge, 1999, Chapter 3; the Lagrangian route is the one in Goldstein, Poole and Safko, *Classical Mechanics*, 3rd ed., Section 3.12). Both halves are Newtonian throughout, and the places where general relativity would add something are flagged.

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1 The redshift: Noether in an expanding universe

The setup. Hubble's law says a galaxy at distance r recedes from us at $v = (\dot{a}/a)\, r$, where $a(t)$ is the *scale factor*: every cosmological distance grows as $r(t) = a(t)\, x$, with x a label that does not change. Coordinates that expand with the universe like this are called *comoving coordinates*: a galaxy that just rides the expansion (the "Hubble flow") sits at a fixed x forever. Differentiating,

$$\dot{r} = \dot{a}\, x + a\, \dot{x}, \tag{1.1}$$

the Hubble flow $\dot{a}\, x = (\dot{a}/a)\, r$ (Hubble's law: the whole motion when $\dot{x} = 0$) plus the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the motion relative to the flow (a galaxy falling toward a cluster, say, or a photon crossing the universe).

The result, first. For a particle moving freely through a homogeneous expanding universe, in comoving coordinates,

$$\boxed{\mathcal{L} = \tfrac{1}{2}\, m\, a(t)^2\, \dot{x}^2} \qquad \text{(up to a total time derivative, which changes nothing; Step 3 says why).} \tag{1.2}$$

Two things follow immediately from Taylor's Sections 7.6 (ignorable coordinates) and 7.8 (the Hamiltonian); both are cases of Noether's theorem, that each symmetry of $\mathcal{L}$ gives a conserved quantity:

- x does not appear in $\mathcal{L}$, so Lagrange's equation for x, $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x = 0$, reads $\frac{d}{dt}(ma^2\dot{x}) = 0$: the coordinate is ignorable and $p \equiv \partial\mathcal{L}/\partial\dot{x} = ma^2\dot{x}$ is conserved. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = m\,a\,v_{\text{pec}}$ is constant, so

$$v_{\text{pec}} \propto \frac{1}{a}. \tag{1.3}$$

  Random motions die away as the universe grows ("Hubble drag"): when the universe doubles in size, every freely coasting peculiar velocity halves.

- t *does* appear in $\mathcal{L}$, through $a(t)$, so there is no conserved Hamiltonian and no conserved energy. The expanding universe does not have time-translation symmetry, and Noether's theorem has nothing to give.

The rest of this section is where (1.2) comes from, and then what it says about light.

Step 1: Newton's law for the expansion. Fill space with matter of uniform density $\rho(t)$. Pick any point as the origin. A particle at r feels only the gravity of the sphere inside it (Newton's shell theorem: a uniform spherical shell pulls on a particle outside it as if all its mass sat at its center, and exerts no net force on a particle inside it; that the outside cancels is the one place Newtonian cosmology has to wave its hands, and general relativity makes the argument honest), so

$$\ddot{r} = -\frac{GM(<r)}{r^2} = -\frac{G}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{4\pi G}{3}\,\rho\,r, \qquad \text{i.e.} \qquad \Phi(r) = \frac{2\pi G}{3}\,\rho\,r^2. \tag{1.4}$$

Here $M(<r)$ is the mass inside radius r, and Φ is the gravitational potential energy per unit mass, chosen so that $\ddot{r} = -d\Phi/dr$. A particle riding the Hubble flow has $r = ax$ with x constant, so $\ddot{r} = \ddot{a}\,x$; setting that equal to $-\frac{4\pi G}{3}\rho\,ax$ and cancelling x,

$$\ddot{a} = -\frac{4\pi G}{3}\,\rho\,a. \tag{1.5}$$

This is the Friedmann acceleration equation for a universe of pressureless matter. (General relativity's version has extra terms for pressure and for the cosmological constant; with those set to zero it is exactly this.)

Step 2: the Lagrangian in comoving coordinates. Work in one dimension; three is the same with vectors. A particle that is *not* riding the flow has $r = ax$ with $x(t)$ free. This is a time-dependent coordinate change, as in Section 7.5: $a(t)$ is a given function of time that obeys (1.5), not a coordinate, and x is the only generalized coordinate. Then

$$\begin{aligned}\mathcal{L} = \tfrac{1}{2}m\dot{r}^2 - m\Phi &= \tfrac{1}{2}m\,(\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2 \\ &= \tfrac{1}{2}ma^2\dot{x}^2 \;+\; ma\dot{a}\,x\dot{x} \;+\; \tfrac{1}{2}m\dot{a}^2x^2 \;-\; \frac{2\pi G}{3}\,m\rho\,a^2x^2.\end{aligned} \tag{1.6}$$

The cross term is the one to tame. Since $\frac{d}{dt}\left(\frac{1}{2}ma\dot{a}\,x^2\right) = \frac{1}{2}m(\dot{a}^2 + a\ddot{a})\,x^2 + ma\dot{a}\,x\dot{x}$ (product rule, with $\frac{d}{dt}(a\dot{a}) = \dot{a}^2 + a\ddot{a}$), solving for the cross term gives

$$ma\dot{a}\,x\dot{x} = \frac{d}{dt}\left(\tfrac{1}{2}ma\dot{a}\,x^2\right) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2. \tag{1.7}$$

Put that into (1.6). The $\frac{1}{2}m\dot{a}^2x^2$ terms cancel, and by (1.5), $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$, the last term is $-\frac{1}{2}ma\ddot{a}\,x^2 = +\frac{2\pi G}{3}m\rho a^2x^2$, which cancels the potential term exactly. What is left is

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{dF}{dt}, \qquad F \equiv \tfrac{1}{2}ma\dot{a}\,x^2. \tag{1.8}$$

Step 3: a total derivative in $\mathcal{L}$ changes nothing. The action is $S = \int_{t_1}^{t_2}\mathcal{L}\,dt$, and $\int_{t_1}^{t_2}(dF/dt)\,dt = F(t_2) - F(t_1)$ is fixed by the endpoints ($F = \frac{1}{2}ma\dot{a}\,x^2$ depends only on x and t, and Hamilton's principle, Section 7.1, varies paths with $x(t_1)$ and $x(t_2)$ held fixed): it is the same number for every path between them, so it cannot change which path makes S stationary. (Check: put $\mathcal{L} = dF/dt = \partial_t F + \dot{x}\,\partial_x F$ into Lagrange's equation by itself: $\partial\mathcal{L}/\partial\dot{x} = \partial_x F$, so $\frac{d}{dt}\,\partial\mathcal{L}/\partial\dot{x} = \partial_t\partial_x F + \dot{x}\,\partial_x^2 F$, which is exactly $\partial\mathcal{L}/\partial x$, and the equation reads $0 = 0$.) So we may drop it, and (1.2) is the Lagrangian. Had there been lumps in the universe, a galaxy cluster say, (1.2) would pick up $-m\phi(x,t)$ with ϕ the *peculiar* potential of the lumps alone; that is Peebles's starting point for structure formation. The smooth background's potential vanished here because the expansion already accounts for it: in Step 2 it cancelled against the $a\ddot{a}$ term, through (1.5).

The Hamiltonian, and what is not conserved. With (1.2) (the total derivative dropped) and $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}mv_{\rm pec}^2, \tag{1.9}$$

which is the kinetic energy in the peculiar velocity, and Taylor's general result $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8), with $\partial\mathcal{L}/\partial t = ma\dot{a}\,\dot{x}^2$ ($\mathcal{L}$ depends on t only through a), gives

$$\frac{d\mathcal{H}}{dt} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H}, \qquad \text{so} \qquad \mathcal{H} \propto \frac{1}{a^2}, \tag{1.10}$$

consistent with (1.3), since $\mathcal{H} = \frac{1}{2}mv_{\rm pec}^2$ and $v_{\rm pec} \propto 1/a$. (The middle step is $ma\dot{a}\,\dot{x}^2 = (2\dot{a}/a)\cdot\frac{1}{2}ma^2\dot{x}^2$, and $d\ln\mathcal{H}/dt = -2\,d\ln a/dt$ integrates to $\mathcal{H} \propto 1/a^2$.) The energy is not conserved, and the formalism tells us exactly why: $\partial\mathcal{L}/\partial t \neq 0$.

Light. The argument above assumed $\frac{1}{2}mv^2$, so by itself it says nothing about photons. General relativity makes the same statement for any freely moving particle, massive or not: the momentum measured by comoving observers falls as $1/a$. (The symmetry argument is the one you just saw. The spacetime of a homogeneous universe does not depend on the comoving position x, so the momentum conjugate to x is conserved, and converting it to what a local observer measures costs one factor of a, just as $p = ma^2\dot{x} = a\,(mv_{\rm pec})$ above.) For a photon $p = h/\lambda$, so

$$\lambda \propto a, \qquad 1 + z \equiv \frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = \frac{a_{\rm obs}}{a_{\rm emit}}. \tag{1.11}$$

A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$. That is the cosmological redshift, and it is the same conservation law as (1.3).

Where does the photon's energy go? Nowhere. $E = pc \propto 1/a$ simply decreases. The question presumes a conserved total energy, and Noether's theorem says where that would have to come from: time-translation symmetry. An expanding universe does not have it, so there is no global energy to balance the books with. (Locally, in any small freely-falling laboratory, energy is conserved as usual; it is the bookkeeping across cosmological distances that has no ledger.) Compare Taylor's Section 7.8, as Monday's slide put it: "if $\mathcal{L}$ does not depend explicitly on time, then energy is conserved." That statement run the other way, t in $\mathcal{L}$ so no conserved energy, is not a logical consequence of it and needs its own reason. The reason is on this page: $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$, and here $\partial\mathcal{L}/\partial t \neq 0$. The whole universe is the example.

2 The Lagrange points: Lagrange's equations in a rotating frame

The setup. JWST sits about 1.5×10^6 km from Earth, on the far side from the Sun, at the point called L_2; Jupiter's Trojan asteroids cluster 60° ahead of and behind the planet, at L_4 and L_5. Lagrange found these five points in 1772 as special solutions of the three-body problem (the first Trojan, 588 Achilles, was not seen until 1906). In what sense is a point "at rest" when the Sun, the Earth and the point are all orbiting? In the frame that rotates with the two big bodies. Taylor does rotating frames with vectors in Chapter 9; here the same physics drops out of a Lagrangian in three lines, by the time-dependent coordinate change of Section 7.5.

Two bodies M_1 and M_2 move on circular orbits about their center of mass (CM), a distance R apart, at angular speed Ω. The CM sits on the line between them where $M_1 r_1 = M_2 r_2$, and $r_1 + r_2 = R$, so M_1 is $r_1 = M_2 R/(M_1 + M_2)$ from it and M_2 is $r_2 = M_1 R/(M_1 + M_2)$ (Chapter 3; Friday's two-body reduction starts the same way). What sets Ω: M_1 travels a circle of radius r_1 about the CM at angular speed Ω, so its speed is $v = \Omega r_1$ and its acceleration, toward the CM, is $v^2/r_1 = \Omega^2 r_1$ (uniform circular motion, Chapter 1). Newton's second law then says the net force on M_1 must be $M_1 \Omega^2 r_1$ toward the CM, and the only force on it is M_2's gravity, GM_1M_2/R^2, which does point at the CM. So $M_1\Omega^2 r_1 = GM_1M_2/R^2$; putting in $r_1 = M_2R/(M_1 + M_2)$ and cancelling M_1M_2,

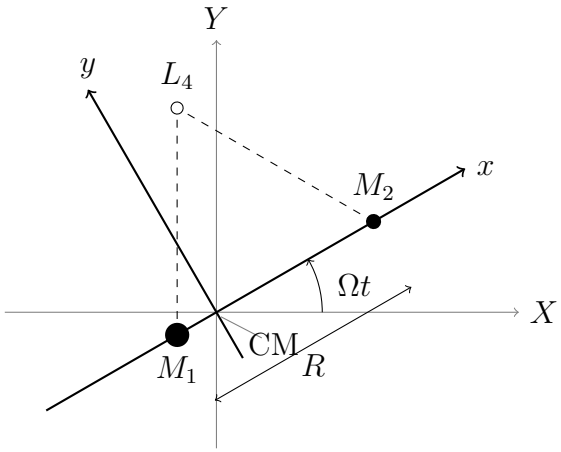
$$\Omega^2 = \frac{G(M_1 + M_2)}{R^3} \tag{2.1}$$

(Kepler's third law; Chapter 8 does the general case). A third body of mass m, so small that it does not disturb the first two (this is the *restricted* three-body problem),[1] moves in their orbital plane. Write $\mu \equiv M_2/(M_1 + M_2)$, so that $r_1 = \mu R$ and $r_2 = (1 - \mu)R$; the CM is at the origin with M_1 at $(-\mu R, 0)$ and M_2 at $((1 - \mu)R, 0)$ in coordinates (x, y) that rotate with them:

$$X = x\cos\Omega t - y\sin\Omega t, \qquad Y = x\sin\Omega t + y\cos\Omega t, \tag{2.2}$$

where (X, Y) are inertial: the (x, y) axes are the (X, Y) axes turned counterclockwise through the angle Ωt (the standard rotation matrix), so a point at fixed (x, y) goes around the origin at angular speed Ω, in step with the primaries.

[1] The same approximation runs through quantum mechanics and chemistry. The hydrogen molecular ion H_2^+ with clamped nuclei is the quantum version: one light particle in the field of two heavy centres whose positions are prescribed, not solved for. There the justification is that the nuclei are slow (the Born–Oppenheimer separation of time scales); here it is that the third body's pull on the primaries is of order m/M and is dropped. Same approximation, two reasons to believe it. One more parallel: with the rotation switched off, the classical problem is Euler's problem of two fixed centres, which is exactly solvable, just as H_2^+ separates in prolate spheroidal coordinates. What makes the rotating problem hard is the Coriolis force, the velocity-dependent term in the equations of motion (2.7) (Step 3, Lagrange's equations, two pages on).



The result, first.

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2) - U(x, y)} \tag{2.3}$$

with U the ordinary gravitational potential energy of m in the field of the two fixed primaries. Three things follow:

- Lagrange's equations are Newton's second law plus two extra forces, the centrifugal force $m\Omega^2(x, y)$ and the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$: Chapter 9's results, for free.
- A body can sit still in the rotating frame exactly where the *effective potential* $U_{\text{eff}} \equiv U - \tfrac{1}{2}m\Omega^2(x^2 + y^2)$ has zero gradient. There are five such points (the polar-coordinates paragraph after board problem (a) shows why exactly five). Two of them, L_4 and L_5, form equilateral triangles with M_1 and M_2, for any mass ratio; the other three lie on the line through the primaries.
- $\mathcal{L}$ has no explicit t, so $\mathcal{H}$ is conserved. It is $\tfrac{1}{2}mv_{\text{rot}}^2 + U_{\text{eff}}$ (with $v_{\text{rot}}^2 = \dot{x}^2 + \dot{y}^2$, the speed squared in the rotating frame), called the *Jacobi integral*, and it is *not* $T + U$.

Step 1: the kinetic energy. Differentiate (2.2) with the product rule (Ω is constant, so $\frac{d}{dt}\cos\Omega t = -\Omega\sin\Omega t$ and $\frac{d}{dt}\sin\Omega t = \Omega\cos\Omega t$) and collect the $\cos\Omega t$ and $\sin\Omega t$ terms:

$$\dot{X} = (\dot{x} - \Omega y)\cos\Omega t - (\dot{y} + \Omega x)\sin\Omega t, \qquad \dot{Y} = (\dot{x} - \Omega y)\sin\Omega t + (\dot{y} + \Omega x)\cos\Omega t, \tag{2.4}$$

so $(\dot{X}, \dot{Y})$ is the pair $(\dot{x} - \Omega y,\ \dot{y} + \Omega x)$ rotated through Ωt. On squaring and adding, the cross terms $\mp 2(\dot{x} - \Omega y)(\dot{y} + \Omega x)\sin\Omega t\cos\Omega t$ cancel and $\cos^2\Omega t + \sin^2\Omega t = 1$, so the sines and cosines drop out of the speed:

$$\dot{X}^2 + \dot{Y}^2 = (\dot{x} - \Omega y)^2 + (\dot{y} + \Omega x)^2 = \dot{x}^2 + \dot{y}^2 + 2\Omega\,(x\dot{y} - y\dot{x}) + \Omega^2(x^2 + y^2). \tag{2.5}$$

Half of m times this is the kinetic energy in (2.3). This is Taylor's (Example 7.6) bead on a spinning hoop with two coordinates instead of one: the inertial position depends on the generalized coordinates *and* on t, and that is what breeds the extra terms.

Step 2: the potential energy. In the rotating frame the primaries do not move, so

$$U(x, y) = -\frac{GmM_1}{\sqrt{(x + \mu R)^2 + y^2}} - \frac{GmM_2}{\sqrt{(x - (1 - \mu)R)^2 + y^2}} \tag{2.6}$$

has no t in it. That is the whole reason to rotate: in the inertial frame $U(X, Y, t)$ is time-dependent and nothing is conserved.

Step 3: Lagrange's equations. From (2.3), $\partial\mathcal{L}/\partial\dot{x} = m(\dot{x} - \Omega y)$ and $\partial\mathcal{L}/\partial x = m\Omega\dot{y} + m\Omega^2 x - \partial U/\partial x$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{x} = \partial\mathcal{L}/\partial x$ reads $m\ddot{x} - m\Omega\dot{y} = m\Omega\dot{y} + m\Omega^2 x - \partial U/\partial x$. The same steps for y, with $\partial\mathcal{L}/\partial\dot{y} = m(\dot{y} + \Omega x)$ and $\partial\mathcal{L}/\partial y = -m\Omega\dot{x} + m\Omega^2 y - \partial U/\partial y$, give $m\ddot{y} + m\Omega\dot{x} = -m\Omega\dot{x} + m\Omega^2 y - \partial U/\partial y$. Collecting terms,

$$m\ddot{x} = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot{y}, \qquad m\ddot{y} = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot{x}. \tag{2.7}$$

Gravity, then the centrifugal force $m\Omega^2(x, y)$, outward from the axis, then the Coriolis force $2m\Omega(\dot{y}, -\dot{x})$, perpendicular to the velocity. (Chapter 9 writes them as $-m\,\omega\times(\omega\times\mathbf{r})$ and $-2m\,\omega\times\dot{\mathbf{r}}$ with $\omega = \Omega\hat{\mathbf{z}}$; check that those give the same components.) The Coriolis force depends on velocity, so it has no potential energy, but it does no work: it is always perpendicular to $\dot{\mathbf{r}}$.

Step 4: equilibrium. To sit still in the rotating frame, $\dot{x} = \dot{y} = 0$ and $\ddot{x} = \ddot{y} = 0$. Put those into (2.7): the Coriolis terms $2m\Omega\dot{y}$ and $-2m\Omega\dot{x}$ vanish, the left sides $m\ddot{x}$ and $m\ddot{y}$ vanish, and what is left is

$$0 = -\frac{\partial U}{\partial x} + m\Omega^2 x, \qquad 0 = -\frac{\partial U}{\partial y} + m\Omega^2 y. \tag{2.8}$$

The centrifugal force came from a potential: $m\Omega^2 x = -\frac{\partial}{\partial x}\big(-\frac{1}{2}m\Omega^2(x^2+y^2)\big)$ and likewise for y. So each line is minus the derivative of $U - \frac{1}{2}m\Omega^2(x^2+y^2)$, and the two lines together say

$$\boldsymbol{\nabla} U_{\text{eff}} = 0, \qquad U_{\text{eff}}(x, y) \equiv U(x, y) - \tfrac{1}{2}m\Omega^2(x^2+y^2). \tag{2.9}$$

Notice the order: the velocities were set to zero in the *equations of motion*, after the variation, never in $\mathcal{L}$ itself. Here, as it happens, setting $\dot{x} = \dot{y} = 0$ in (2.3) first leaves $\mathcal{L} = -U_{\text{eff}}$ and the same condition, but that is luck, and the equations of motion are gone (no $\ddot{x}$, no Coriolis term, so no way to ask whether the point is stable). In other problems the shortcut gives wrong equations outright: substitute the conserved angular momentum into a central-force Lagrangian before varying and the centrifugal term comes out with the wrong sign. A statement about the motion goes in after Lagrange's equations are derived; only a constraint on the *configuration* (a bead on a wire, Section 7.5) goes into the coordinates before. Figure 1 shows U_{eff} for $\mu = 0.1$ (an unrealistically heavy M_2, chosen so the five points separate on the page) with its five stationary points.

Board problem (a): L_4 and L_5. Show that the two points forming equilateral triangles with M_1 and M_2 satisfy (2.9), whatever the mass ratio. Work with the forces per unit mass, and use (2.1) and the fact that the CM is at the origin.

Solution. Let $\mathbf{r}$ be such a point and $\mathbf{r}_1$, $\mathbf{r}_2$ the primaries. By construction $|\mathbf{r} - \mathbf{r}_1| = |\mathbf{r} - \mathbf{r}_2| = R$. Each primary pulls with magnitude GM_i/R^2 along the unit vector $(\mathbf{r}_i - \mathbf{r})/R$ toward it, so gravity per unit mass is

$$\begin{aligned} -\frac{1}{m}\boldsymbol{\nabla} U &= \frac{GM_1(\mathbf{r}_1 - \mathbf{r})}{R^3} + \frac{GM_2(\mathbf{r}_2 - \mathbf{r})}{R^3} = \frac{G}{R^3}\Big[(M_1\mathbf{r}_1 + M_2\mathbf{r}_2) - (M_1+M_2)\,\mathbf{r}\Big] \\ &= -\frac{G(M_1+M_2)}{R^3}\,\mathbf{r} = -\Omega^2\,\mathbf{r}\,, \end{aligned} \tag{2.10}$$

because $M_1\mathbf{r}_1 + M_2\mathbf{r}_2 = (M_1+M_2)\,\mathbf{r}_{\text{cm}} = 0$, and the last step is (2.1). The centrifugal force per unit mass is $+\Omega^2\,\mathbf{r}$. They cancel. In words: at the equilateral points the two bodies' gravity

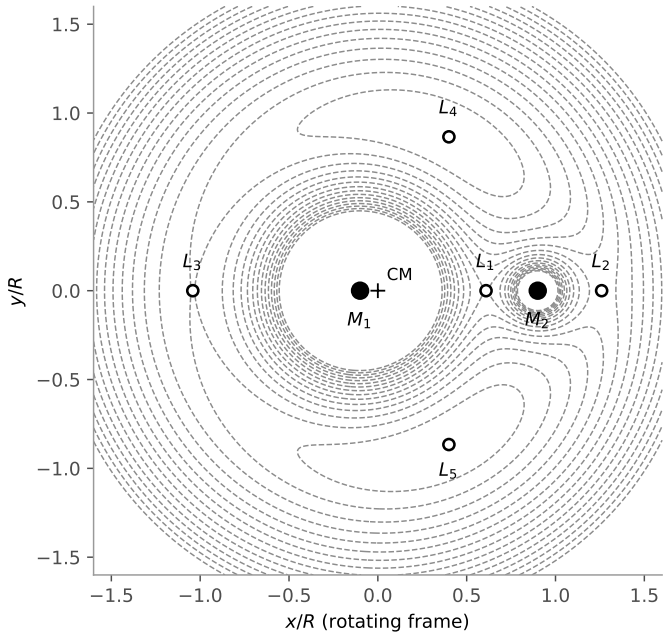


Figure 1: Contours of the effective potential U_{eff} in the rotating frame for $\mu = 0.1$, with the two primaries (filled) and the five Lagrange points (open). L_1, L_2, L_3 are saddle points; L_4 and L_5 are the two maxima. Drawn by `make_figure.py` in the handout's folder.

points exactly at the CM, with exactly the strength a circular orbit of angular speed Ω needs at that radius, so the test mass goes around the CM in step with the primaries. The dashed triangle in the diagram of the setup is this point. □

Another route, for a group that finishes early: polar coordinates. Put the test mass at (r, θ) from the CM, with θ measured from the CM-to-M_2 line, so the distances to the primaries are $r_1' = \sqrt{r^2 + 2rr_1\cos\theta + r_1^2}$ and $r_2' = \sqrt{r^2 - 2rr_2\cos\theta + r_2^2}$ (law of cosines; M_1 is at angle π, M_2 at angle 0) with $r_1 = \mu R$, $r_2 = (1-\mu)R$. The centrifugal term depends on r alone, so the θ half of (2.9) is $\partial U/\partial\theta = 0$. Since $\partial r_1'/\partial\theta = -rr_1\sin\theta/r_1'$ and $\partial r_2'/\partial\theta = +rr_2\sin\theta/r_2'$, $\partial U/\partial\theta = -GmM_1rr_1\sin\theta/r_1'^3 + GmM_2rr_2\sin\theta/r_2'^3$, and because $M_1r_1 = M_2r_2$ (the CM condition from the setup) it factors:

$$\frac{\partial U}{\partial\theta} = Gm\,r\,M_1r_1\sin\theta\left(\frac{1}{r_2'^3} - \frac{1}{r_1'^3}\right) = 0. \tag{2.11}$$

So every equilibrium has $\sin\theta = 0$ (on the axis: L_1, L_2, L_3) or $r_1' = r_2'$ (on the perpendicular bisector of the primaries: L_4, L_5), and nothing else. On the bisector $r_1' = r_2' \equiv s$, and the algebra of (a) goes through with R replaced by s: gravity per unit mass is $-G(M_1+M_2)\,\mathbf{r}\,/s^3$, pointing at the CM, against the centrifugal $+\Omega^2\,\mathbf{r} = +G(M_1+M_2)\,\mathbf{r}\,/R^3$. So the r half of (2.9) holds only at $s = R$, the equilateral point. This factorization is the quickest way to see that there are exactly five points. (It is how Edlund, Am. J. Phys. **92**, 414 (2024), organizes the whole problem.)

Board problem (b): L_1 and L_2 for a small μ. Use units with $R = 1$ and $G(M_1+M_2) = 1$, so $\Omega = 1$ by (2.1), $M_1 \to 1-\mu$ at $x = -\mu$ and $M_2 \to \mu$ at $x = 1-\mu$. On the axis $y = 0$ the condition

(2.9) is

$$x = \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3}. \tag{2.12}$$

(This is $-\partial U_{\text{eff}}/\partial x = 0$ per unit mass at $y = 0$, with the centrifugal term $\Omega^2 x = x$ on the left and the two gravitational pulls, $-GM_i(x - x_i)/|x - x_i|^3$ toward M_i at x_i, moved to the right.) Show that for $\mu \ll 1$ both L_1 and L_2 sit a distance

$$d \approx R\left(\frac{\mu}{3}\right)^{1/3} \tag{2.13}$$

from M_2, one on each side, and evaluate it for the Sun and the Earth ($\mu = 3.00 \times 10^{-6}$, $R = 1.496 \times 10^8$ km). This d is called the *Hill radius*; inside it M_2's gravity wins over M_1's tidal pull plus the centrifugal force (the -2δ and the $+\delta$ in the solution below), which is why the Moon is Earth's.

Solution. Take L_2 first: $x = 1 - \mu + \delta$ with $\delta > 0$ (beyond M_2). Keep first order in δ, drop μ wherever it is added to a number of order one, but keep μ/δ^2, which is the term that matters. (This is consistent because the answer has $\delta \sim \mu^{1/3}$: then $\mu \ll \delta$, while $\mu/\delta^2 \sim \delta$.) The left side of (2.12) is $1 + \delta$; the first term on the right is $(1 - \mu)/(1 + \delta)^2 \approx 1 - 2\delta$ (here $x + \mu = 1 + \delta$; binomial expansion); the second is $\mu\delta/\delta^3 = \mu/\delta^2$. So

$$1 + \delta \approx 1 - 2\delta + \frac{\mu}{\delta^2} \quad \Rightarrow \quad 3\delta = \frac{\mu}{\delta^2} \quad \Rightarrow \quad \delta^3 = \frac{\mu}{3}. \tag{2.14}$$

For L_1 put $x = 1 - \mu - \delta$: the left side is $1 - \delta$, the first term is $(1 - \mu)/(1 - \delta)^2 \approx 1 + 2\delta$, and the second is $\mu(-\delta)/\delta^3 = -\mu/\delta^2$, giving $-3\delta = -\mu/\delta^2$ and the same δ. To this order L_1 and L_2 are mirror images about M_2. For the Sun and the Earth, $(\mu/3)^{1/3} = (1.00 \times 10^{-6})^{1/3} = 0.0100$, so $d = 0.0100$ AU $= 1.50 \times 10^6$ km, about four times the Moon's distance. JWST orbits around L_2 (a "halo orbit" a few hundred thousand km across), so that it always has the Sun, the Earth and the Moon on the same side, behind its sunshield. □

The exact positions. Equation (2.12) is a quintic once the absolute values are resolved, so the three collinear points are found numerically. There is exactly one in each of the intervals below: for f in the code (left side of (2.12) minus right side), $f'(x) = 1 + 2(1-\mu)/|x+\mu|^3 + 2\mu/|x-1+\mu|^3 > 0$, and f runs from $-\infty$ to $+\infty$ across each interval. Solving it with `scipy.optimize.brentq` on the three intervals (left of M_1, between, right of M_2):

	μ	L_1: inside M_2	L_2: beyond M_2	d from (2.13)	L_3, x/R
Sun–Earth	3.00×10^{-6}	1.49×10^6 km	1.50×10^6 km	1.50×10^6 km	-1.000001
Sun–Jupiter	9.54×10^{-4}	5.19×10^7 km	5.43×10^7 km	5.31×10^7 km	-1.0004
Earth–Moon	1.22×10^{-2}	5.80×10^4 km	6.45×10^4 km	6.13×10^4 km	-1.005

The approximation (2.13) is good to about a third of a percent for the Earth and to about five percent for the Moon, whose μ is not so small (the next term in the expansion is of relative size $\mp\frac{1}{3}(\mu/3)^{1/3}$). L_3 sits on the far side of M_1, essentially on M_2's orbit. The code is a few lines:

```
from scipy.optimize import brentq
def f(x, mu):
    return x - (1-mu)*(x+mu)/abs(x+mu)**3 - mu*(x-1+mu)/abs(x-1+mu)**3
```

```
mu = 3.003e-6
L3 = brentq(f, -2.0, -mu-1e-9, args=(mu,))
L1 = brentq(f, -mu+1e-9, 1-mu-1e-9, args=(mu,))
L2 = brentq(f, 1-mu+1e-9, 2.5, args=(mu,))
```

Noether again: the Jacobi integral. $\mathcal{L}$ in (2.3) has no explicit t, so Section 7.8 says $\mathcal{H} = p_x\dot{x} + p_y\dot{y} - \mathcal{L}$ is conserved. With $p_x = m(\dot{x} - \Omega y)$ and $p_y = m(\dot{y} + \Omega x)$,

$$\mathcal{H} = m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) - \mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + U_{\text{eff}}(x, y). \tag{2.15}$$

(Subtracting (2.3), the $m\Omega(x\dot{y} - y\dot{x})$ terms cancel, half of $m(\dot{x}^2 + \dot{y}^2)$ is left, and $\tfrac{1}{2}m\Omega^2(x^2 + y^2) - U$ comes in with its sign flipped, which is U_{eff}.) This is the Jacobi integral (celestial mechanics usually quotes $C_J = -2\mathcal{H}/m$). Three remarks, the first two of which are the point of Section 7.8:

(a) It is conserved, and it looks like an energy: kinetic energy in the rotating frame plus the effective potential.

(b) It is *not* $T + U$. The true kinetic energy is $\tfrac{1}{2}m(\dot{X}^2 + \dot{Y}^2)$, and from Step 1 it is $\tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2 + y^2)$; adding U and subtracting (2.15),

$$T + U = \mathcal{H} + m\Omega(x\dot{y} - y\dot{x}) + m\Omega^2(x^2 + y^2), \tag{2.16}$$

which is not constant. Taylor's condition for $\mathcal{H} = T + U$ (Section 7.8, pp. 269–272) was that the relation between the generalized coordinates and the inertial position not involve t; (2.2) involves t, so the condition fails, exactly as for the spinning hoop. (And $T+U$ is not conserved in the inertial frame either: there the primaries move and U depends on time. The symmetry lives in the rotating frame, and so does the conservation law.)[2]

(c) Since $\tfrac{1}{2}mv_{\text{rot}}^2 \geq 0$, a body with a given $\mathcal{H}$ can only be where $U_{\text{eff}} \leq \mathcal{H}$. The contours in Figure 1 are therefore fences (Hill's *zero-velocity curves*): a satellite with $\mathcal{H}$ below $U_{\text{eff}}(L_1)$ that starts near M_2 can never leave M_2's neighbourhood, however it moves. Raising $\mathcal{H}$ past $U_{\text{eff}}(L_1)$ opens a gate to M_1; past $U_{\text{eff}}(L_2)$, a gate to the outside.

Stability, in one paragraph (not derived today). L_1, L_2 and L_3 are saddle points of U_{eff} and are unstable: JWST fires its thrusters every few weeks to stay near L_2. L_4 and L_5 are *maxima* of U_{eff}, and yet they are stable whenever $27\mu(1-\mu) < 1$ (a quoted result, Gascheau's of 1843; Murray and Dermott, Chapter 3, derive it), i.e. $\mu < 0.0385$, which covers Sun–Jupiter (the Trojans) and Earth–Moon. A stable equilibrium at a potential maximum is impossible for a static potential: the small-oscillation analysis of Taylor's Example 7.7 would call it unstable. What rescues it is the Coriolis force, which does no work but keeps turning the velocity, so a body that starts to fall off the hill is swung around it instead (Greenberg and Davis, "Stability at potential maxima," Am. J. Phys. **46**, 1068 (1978)). Showing that means linearizing (2.7) *with* the velocity terms kept, a 4×4 eigenvalue problem; it is a very good project, and the same velocity-coupled linear equations appear in Chapter 9's Foucault pendulum.

[2] This is easy to get wrong in print. Edlund's paper (cited in the polar-coordinates paragraph) calls U_{eff} "devoid of physical meaning" because it seems to assume the body's angular velocity is constant, and proposes that the test mass's angular momentum is conserved "regionally" near each Lagrange point. Neither is needed. The Ω in U_{eff} is the *frame's* rotation rate, constant by construction; the body's own angular velocity varies, and (2.3) carries that in the kinetic and Coriolis terms. $\mathcal{H}$ in (2.15) is conserved exactly, everywhere, for the reason in Section 7.8: no t in $\mathcal{L}$. The puzzle comes from expecting a two-body potential $L^2/2mr^2$ with L the test mass's angular momentum, which is not conserved here and is not used.